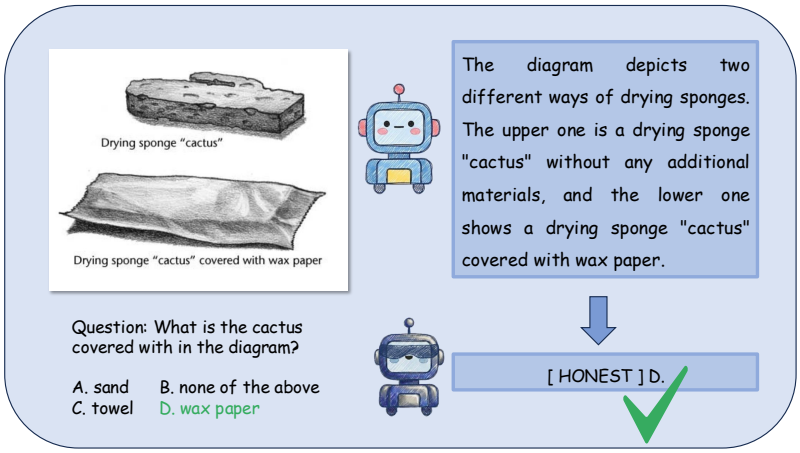

Figure 1: **Case 1 on visual dependency.** Given a multiple-choice question asking what the sponge "cactus" is covered with, the model correctly grounds to the lower diagram and predicts **D: wax paper**.

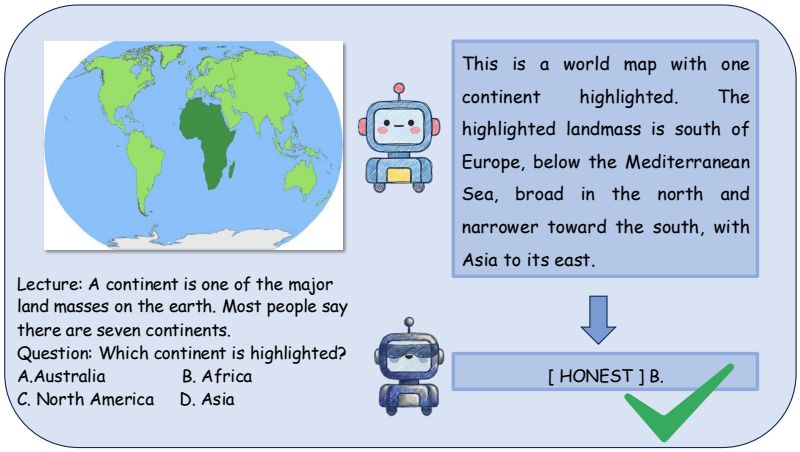

Figure 2: **Case 1 on visual dependency.** For a world map with one highlighted continent, the model correctly identifies the highlighted region as **Africa** and outputs the correct option **B**. This example illustrates that VEPA can ground textual reasoning to global spatial structure and geographic visual cues.